

Fast Gradient Method (FGM) Fine-Tuning for Bengali Sarcasm Detection

Talha Jobayer Zihan

*Department of Computer Science and Engineering
Rajshahi University of Engineering and Technology
Rajshahi-6204, Bangladesh
jobayertalha2020@gmail.com*

Mohiuddin Ahmed

*Department of Computer Science and Engineering
Rajshahi University of Engineering and Technology
Rajshahi-6204, Bangladesh
mohiuddin.nirob.mn@gmail.com*

A.F.M Minhazur Rahman

*Department of Computer Science and Engineering
Rajshahi University of Engineering and Technology
Rajshahi-6204, Bangladesh
m.r.saurov@gmail.com*

Foyez Ahmed Dewan

*Department of Computer Science and Engineering
Rajshahi University of Engineering and Technology
Rajshahi-6204, Bangladesh
foyez.ruet767@gmail.com*

Abstract—Language and expression are the way of expressing human emotions, which help humans express feelings and attitudes. However, sarcasm usually masks the true motive, and thus the interpretation of sentiments in online communication is especially difficult. Sarcasm in Bengali is an extremely common culturally sensitive type of humor that can be easily misunderstood on social networks. To overcome this difficulty, we introduce a complete supervised deep learning system in Bengali sarcasm detection, which is based on the BanglaBERT using Fast Gradient Method (FGM) adversarial fine-tuning, layer selective freezing, and regularization optimization. Our experiments on two benchmark datasets, BenSarc and BanglaSarc3, show that our method provides better and more consistent performance than the current semisupervised GAN-BERT architectures while containing a simpler yet more efficient architecture. The proposed model has an accuracy of 77.71% and a weighted F1 score of 0.777 with 50 percent of the data labeled and up to 89.00% accuracy and 0.890 weighted F1 with 100 percent of the data labeled on the BenSarc dataset. Extended to the ternary BanglaSarc3 data set of bilingual Bangla-English inputs and R-Drop regularization, the precision and macro F1 are 82.74% and 0.8275, respectively. These findings point to the fact that a properly optimized supervised BanglaBERT model, enhanced with both adversarial and consistency losses, can be better than semi-supervised alternatives, which is a strong direction to achieve correct sarcasm perception and sentiment detection in low-resource languages such as Bengali.

Index Terms—Sarcasm detection, Bengali, BenSarc, BanglaSarc3, BanglaBERT, adversarial training, FGM, transformers

I. INTRODUCTION

Language is one of the basic means of human communication where people can express thoughts, emotions, humor, and will. Due to the rapid growth of social media networks, the significance of natural language processing (NLP) methods has increased that can read and extract human sentiment. The problem of sarcasm is to the detection of sentiment and toxicity because of its ambiguous and ironic nature, where the literal meaning of a statement is frequently the opposite of

the intended sentiment. Sarcasm is difficult to detect due to the existence of indirect practical hints, context-dependent expressions, and cultural differences. These issues are increased in the Bengali language through linguistic richness, idiomatic variety, and dearth of annotated resources. [1] [2]

Sentiment analysis, categorized as a Natural Language Processing (NLP) task and the term *sentiment analysis*, also known as *opinion mining*, refers to the process of determining whether an individual's perspective on a particular subject, as expressed in text, leans toward positivity, negativity, or neutrality. Text analysis systematically pulls out meaningful information and semantic understanding from text using NLP techniques. It involves identifying the sentiment conveyed by the writer, which may be classified as positive, negative, or neutral [1].

Datasets such as BanglaSarc [1], BenSarc [2], and BanglaSarc3 [3] have been introduced to facilitate sarcasm detection in the Bengali language. These corpora include sarcastic, non-sarcastic, and neutral remarks collected from social media platforms such as Facebook and YouTube, providing valuable benchmarks for model evaluation.

Early studies relied on traditional feature-based classifiers, such as Support Vector Machines (SVM) and Naïve Bayes, which were limited in their ability to capture contextual nuances. So it has shown that the advancement of deep learning has led to noticeable enhancements in the performance of neural network designs across several natural language processing (NLP) applications, including language modeling, text classification, and machine translation. But despite the advancements in algorithms and methodologies over the past few years, there are still numerous issues in the field that have not yet been solved. These still struggle to capture figurative semantics and long-range dependencies in text. The complexity of sarcasm for understanding is further amplified by contextual ambiguities and generational differences in the way sarcasm is used online, as noted by previous studies [4]

and [5]

To overcome these limitations, at the very core of our approach, we focus on recent advancements in transformer-based models that have brought significant improvements to Bengali NLP by enabling context-sensitive token representations through self-attention mechanisms [6] [7]. This paper proposes an end-to-end supervised deep learning framework for Bengali sarcasm detection based on BanglaBERT [8] fine-tuning with adversarial optimization. Specifically, the approach incorporates the Fast Gradient Method (FGM) [9] for improvement of robustness, selective encoder layer freezing for efficient adaptation, and R-Drop consistency regularization for improvement of generalization in bilingual inputs.

The main contributions of this paper are summarized as follows:

- A supervised Bengali sarcasm detection framework is presented using BanglaBERT fine-tuning with FGM [9] adversarial learning, R-Drop regularization, and layer-freezing strategies.
- Extensive experiments are conducted on two major Bengali sarcasm datasets, BenSarc and BanglaSarc3, covering both binary and ternary classification setups.
- Comparative evaluation against the previous GAN-BERT [10] semi-supervised approach demonstrates the superior accuracy, robustness, and stability of the proposed supervised model.
- The results highlight the importance of monolingual pretraining, adversarial perturbation, and cross-lingual consistency in improving sarcasm comprehension for low-resource languages.

The remainder of this paper is organized as follows: Section II reviews related research on sarcasm detection and sentiment analysis. Section III presents the proposed methodology and datasets. Section IV describes the experimental setup and implementation details. Section V discusses the results and comparative analysis, and Section VI concludes the paper with key findings and future research directions.

II. RELATED WORKS

Works on sarcasm detection in Bengali have grown significantly in recent years, supported by new annotated datasets and advancements in neural language models.

The first major contribution, BanglaSarc [1], introduced around 5,000 social media comments collected from Facebook, YouTube, and blogs, labeled as sarcastic or non-sarcastic. It served as an initial resource for benchmarking classical machine learning algorithms. Subsequently, BenSarc [2] expanded the scale with 25,636 Facebook comments and provided a consistent evaluation framework across machine learning, deep learning, and transfer learning models. This work facilitates further experimentation using pre-trained transformer architectures such as BERT and its multilingual variants.

Further development came from Lora et al. [10], who proposed a semi-supervised *GAN-BERT* a hybrid approach for

Bengali sarcasm detection. Their method leveraged unlabeled data to handle ambiguous or confusing sarcastic expressions, establishing one of the first generative frameworks for this task. To enhance linguistic granularity, BanglaSarc3 [3] was later introduced as a ternary benchmark comprising 12,089 balanced comments across sarcastic, non-sarcastic, and neutral labels. This dataset expressed understanding of delicate humor and pragmatic neutrality in Bangla communication.

Recent comparative analyses [5] explored diverse models including CNN-BiLSTM, logistic regression, and fine-tuned transformer networks, while interpretable approaches [4] applied explainable AI techniques to BERT-based classifiers to reveal decision transparency. However, most prior research has not extensively examined adversarial robustness, cross-layer regularization, or consistency alignment for sarcasm detection in low-resource languages.

Through a comprehensive review of prior work, it has been observed that numerous studies have implemented various machine and deep learning approaches for sarcasm detection in the Bengali language. Despite these efforts, only a few have adequately addressed the challenge of modeling contextual and pragmatic meaning, which is essential for understanding sarcasm in text. Most existing systems struggle with complex or subtly expressed sarcastic expressions, limiting their practical effectiveness in real-world conversational settings.

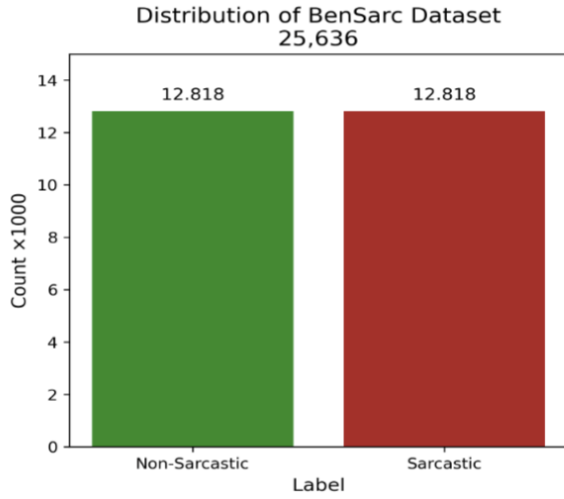
Clearly, detecting textual emotion requires going beyond surface-level text analysis. Which can be achieved by using sarcasm detection approaches. Few studies have explored attention-based or semi-supervised approaches for Bengali sarcasm detection, while most relied on conventional machine or deep learning methods. Only a limited number utilized transformer models what actually works great in this NLP field, whereas our work builds on this by fine-tuning a pre-trained transformer with adversarial learning to enhance performance.

III. METHODOLOGY

A supervised deep learning framework has been developed for Bengali sarcasm detection utilizing the BanglaBERT [8] architecture through fine-tuning and adversarial learning. The overall methodology follows a structured systematic sequence beginning with data preprocessing and preparation, where sarcasm-preserving normalization and cleaning are performed to retain linguistic nuances. Subsequently, subword tokenization is applied to ensure proper handling of the complex morphology of the Bengali language. Then for the next step encoding, fine-tuning is conducted using the integration of the Fast Gradient Method (FGM) [9] to enhance adversarial robustness with BanglaBert [8], while selective layer freezing is employed to stabilize learning and prevent overfitting. We additionally use bilingual inputs and R-Drop. An overview of our proposed methodology is illustrated in Figure 3.

A. Dataset Description

This section provides an overview of the datasets used in this study, highlighting their composition, labeling scheme and

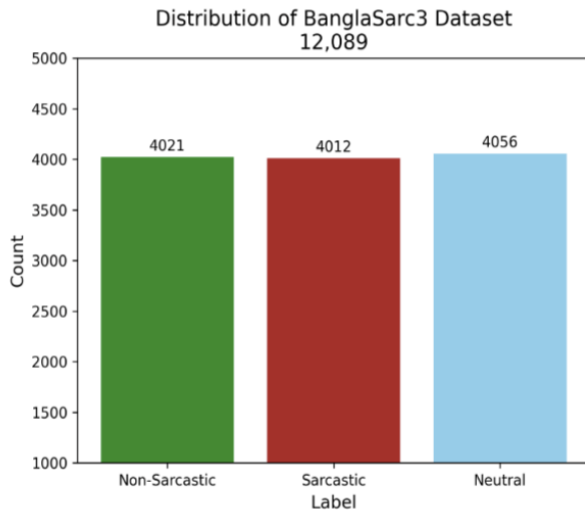


(a) Class distribution

Comments	Polarity
শুধু মাত্র এই পোস্টে কमेंট করার জন্য বাড়ির এক মাত্র ডিম পাড়া ঘোড়া বিক্রি করে এম বি কিনে বসে ছিলাম।	Sarcastic(1)
ফিতরা অর্থ দিয়েও আদায় করা যায়।	Non-Sarcastic(0)

(b) Representative comment samples

Fig. 1: Overview of the BenSarc dataset: (a) shows the distribution of sarcastic and non-sarcastic instances, and (b) presents sample comments.



(a) Class distribution

Bangla Comments	English Translation	Sarcasm Label
ঠাকুর ঘরে কে আমি কলা খাই না	I don't eat bananas in Tagore's house	Sarcastic
পরিবেশ রক্ষায় নিম গাছ সত্যিই অসাধারণ সবাইকে গাছ লাগানো উচিত	Neem tree is really great for protecting the environment. Everyone should plant	Non-Sarcastic
টং দোকানের চা ভালো লাগলে সেখানে যান	Go there if you like Tong shop tea	Neutral

(b) Representative comment samples

Fig. 2: Overview of the BanglaSarc3 dataset: (a) shows the distribution of sarcastic, non-sarcastic, and neutral instances, and (b) presents example comments.

linguistic characteristics. Each dataset is described in detail in the following subsections.

1) BenSarc Dataset

The BenSarc [2] dataset was introduced as a foundational benchmark for Bengali sarcasm detection. It consists of 25,636 Facebook comments, evenly divided into 12,818 sarcastic and 12,818 non-sarcastic instances. The dataset is balanced, containing an equal number of labeled examples across both classes. As illustrated in Figure 1, (a) shows the class distribution, while (b) presents representative samples that capture the linguistic characteristics and sarcastic expressions typical of Bengali social media.

The dataset is evenly distributed, containing an equal number of sarcastic and non-sarcastic labeled evaluations as shown

above. Samples demonstrate the linguistic characteristics and sarcastic patterns found and collected from Bengali social media comments.

2) BanglaSarc3 Dataset

The BanglaSarc3 [3] dataset was introduced as an extended ternary benchmark for Bengali sarcasm detection. It comprises 12,089 Facebook comments categorized into three distinct classes: Sarcastic, Non-Sarcastic, and Neutral. The dataset is fairly balanced across the three categories, as shown in Figure 2, with (a) illustrating the distribution across classes and (b) displaying representative text examples that capture the diverse linguistic patterns and bilingual (Bangla–English) expressions typical of social media sarcasm. BanglaSarc3 provides a valuable resource for multiclass and cross-lingual

sarcasm detection, establishing a comprehensive benchmark for Bengali NLP research.

B. Proposed Methodology

BanglaBERT [8], which is a Bengali based BERT model implementation, was pre-trained on a large Bengali corpus to learn linguistic and syntactic patterns, enabling effective fine-tuning for tasks like sarcasm detection. The overall methodology is shown in Figure 3.

The collected dataset comprises numerous user-generated comments exhibiting diverse linguistic styles, emotions, and contexts. While this diversity adds significance, it also introduces noise that can interrupt model learning. Hence, pre-processing is a vital initial step to clean irrelevant text while preserving pragmatic and emotional cues crucial for sarcasm detection.

For the BenSarc [2] dataset, a 6-tier preprocessing pipeline was applied where URLs, user mentions, and redundant whitespace were removed, but punctuation, emojis, elongations, and repeated characters were retained to preserve expressive cues what is vital for sarcasm detection. Stemming and lemmatization were ignored to maintain morphological richness and stylistic nuances. For BanglaSarc3 [3], which contains bilingual Bangla–English texts, inputs were concatenated in the format: *Bangla*: <bn> [SEP] *English*: <en>, enabling the model to jointly interpret cross-lingual context while minimizing noise and preserving pragmatic signals.

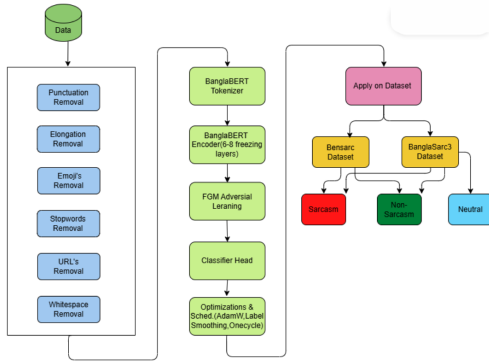


Fig. 3: Architecture of our proposed approach.

After preprocessing, tokenization and embedding generation were performed using the BanglaBERT [8] WordPiece tokenizer with a maximum sequence length of 128. The encoded sequences were fed into BanglaBERT [8] to obtain contextual embeddings. Selective layer freezing was employed to balance adaptation and stability typically freezing the lower six to eight transformer layers, with extended up to ten layers to analyze efficiency–accuracy trade-offs.

To increase BERT model robustness, adversarial learning is incorporated using the Fast Gradient Method (FGM) [9]. In each training batch, adversarial perturbations are introduced to the embedding layer by adding a small, normalized gradient-based noise, forcing the model to generalize better to minor

variations in the input. For the ternary BanglaSarc3 [3] dataset, additional regularization is applied through R-Drop, which introduces a symmetric Kullback–Leibler divergence term between two stochastic forward passes, encouraging consistency and stability in bilingual predictions.

The classification component employs a lightweight multi-layer perceptron (MLP) that processes the [CLS] representation from BanglaBERT [8]. It uses GELU activation, dropout, and a Softmax layer for final label probabilities. Training is optimized with AdamW using discriminative learning rates updating the encoder and classification head at separate rates to balance pre-trained knowledge and task-specific adaptation. Additional advanced techniques such as cosine annealing, OneCycleLR scheduling, label smoothing (0.1), gradient clipping (0.5), and early stopping based on validation F1-score enhance training stability and convergence.

The architecture was evaluated on two benchmark datasets BenSarc [2] and BanglaSarc3 [3] for Bengali sarcasm detection. BenSarc [2] represents a balanced binary setup (sarcastic vs. non-sarcastic), while BanglaSarc3 [3] extends it to a balanced ternary classification with sarcastic, non-sarcastic, and neutral expressions.

The proposed architecture is a fully supervised architecture built upon BanglaBERT [8] fine-tuning with adversarial and regularization strategies. The process begins with 6-tier text preprocessing to preserve the emotion of the text. Then the cleaned data is tokenized and passed through the BanglaBERT [8] encoder, where adversarial noise is introduced via the Fast Gradient Method (FGM) [9] to improve robustness. Selective layer freezing stabilizes learning while retaining pretrained linguistic knowledge, and a lightweight classifier head with dropout and GELU activation performs the final prediction for Bengali sarcasm detection.

IV. EXPERIMENTAL SETUP AND IMPLEMENTATION

The proposed BanglaBERT-based sarcasm detection framework was evaluated on two benchmark datasets: BenSarc [2], BanglaSarc3 [3]. Implementations used PyTorch and Hugging Face Transformers, executed on Kaggle and Google Colab with NVIDIA T4 GPUs (16 GB) and 30 GB RAM.

Two configurations were explored for BenSarc [2]. In the first, the dataset was split into 70%, 10%, and 20% for training, validation, and testing respectively with the training portion further divided into 3%, 30%, 40%, and 50% labeled fractions to evaluate supervision efficiency. The model trained for eight epochs on the 3% and 30% subsets and four epochs on the 40% and 50% subsets using a batch size of 16. In the second configuration, the full dataset (100% labeled) was used with an 80%,10%,10% split and trained for 15 epochs with a batch size 32 and patience based early stopping. Layer-freezing experiments across six, eight, and ten encoder layers revealed that freezing six layers yielded the best balance between efficiency and accuracy.

For BanglaSarc3 [3], the setup adopted a bilingual input format combining Bangla text with its English translation to support cross-lingual learning. The split was 80%,10%,10%

for training, validation, and testing, respectively. It was applied for 15 epochs with a batch size of 32 and patience-based early stopping. The dataset was upsampled to ensure balanced class representation during training. Then applying layer-freezing experiments across six, eight, and ten encoder layers revealed that freezing six layers yielded the best balance between efficiency and accuracy compared to eight and ten layers.

V. EXPERIMENTAL RESULTS AND ANALYSIS

In the next step of our study, the experiments were conducted to evaluate the effectiveness of the proposed supervised BanglaBERT [8] framework combined with Fast Gradient Method (FGM), layer freezing, and R-Drop regularization. Experiments were performed on two major Bengali sarcasm datasets BenSarc [2] and BanglaSarc3 [3] to investigate how supervision, adversarial robustness, and freezing layers affect performance. The aim of our study was to assess whether a supervised approach with adversarial learning models might improve the efficacy of predictions.

A. BenSarc Results

1) Results with Varying Labeled Data (3-50%)

TABLE I: Experimental results of the BenSarc [2] (3-50% labeled data)

Model	3%	30%	40%	50%
Epochs	8	8	4	4
BanglaBERT (with FGM)	68.08	76.44	77.11	77.71

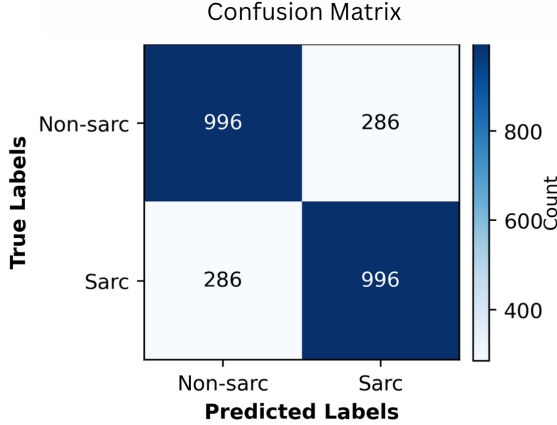


Fig. 4: Confusion matrix for BenSarc [2] (50% labeled data).

The experiments evaluated the effect of different labeled data percentages on BenSarc [2] dataset performance shown in the TABLE I. The accuracy grows consistently with higher labeled supervision, increasing from 68.08% at 3% labels to 77.71% at 50%, marking a 14.16% improvement. The implementation of BanglaBERT [8] with FGM enhanced robustness, particularly under low-resource settings, stabilizing training and preventing overfitting in smaller label regimes. At 50% labeled data, the model achieved 77.71% accuracy, slightly surpassing the semi-supervised GAN-BERT [10], which reported

TABLE II: Experimental results of the BenSarc [2]

(100% labeled data with Layer Freezing)				
Frozen Layers	Parameters	Best Val F1	Test Acc	Test F1
6	42,823,298	0.8981	0.8900	0.8900
8	28,647,554	0.8947	0.8865	0.8865
10	14,471,810	0.8685	0.8623	0.8623

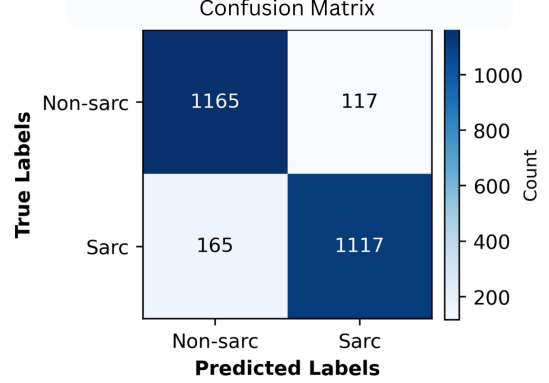


Fig. 5: Confusion matrix for BenSarc [2] (100% data).

77.1%. This result was obtained without using unlabeled samples, demonstrating the effectiveness of the supervised fine-tuning strategy with FGM. The confusion matrix in Fig. 4 shows balanced classification performance across sarcastic and non-sarcastic classes using fully supervised 50% data of the dataset.

2) Results with 100% Labeled Data and Layer Freezing

Experiments with fully supervised data examined with different layer-freezing strategies shown in the TABLE II. Freezing the first six layers yielded the best outcome with 89.00% accuracy and 0.890 F1-score. Freezing eight layers slightly reduced accuracy to 88.65%, while ten-layer freezing dropped it to 86.23%. Increasing freezing layers caused gradual performance degradation, with the six-layer setup outperforming the eight- and ten-layer configurations by 2.35% and 6.77%, respectively.

This indicates that moderate layer freezing preserves pre-trained representations while allowing effective task-specific adaptation. The six-layer configuration improved test accuracy by approximately 2.35% over eight frozen layers and 6.77% over ten frozen layers.

The confusion matrix in Fig. 5 shows balanced predictions across both classes, confirming the model's ability to capture subtle sarcastic nuances.

B. BanglaSarc3 Results

1) Results with Layer Freezing on Ternary Classification

The BanglaSarc3 dataset experiments evaluated layer freezing strategies for ternary classification shown in the TABLE III. The best outcome was achieved by freezing six layers, yielding 82.74% accuracy and 0.8275 macro F1-score. Increasing the freeze depth to eight and ten layers reduced

accuracy by 2.54% and 14.38% respectively, reflecting similar degradation trends.

TABLE III: Experimental Results for the BanglaSarc3 [3]

(Ternary)				
Frozen Layers	Trainable Params	Best Val F1	Test Acc	Macro F1
6	42,823,683	0.8305	0.8274	0.8275
8	28,647,939	0.8072	0.8020	0.8007
10	14,472,195	0.6873	0.6836	0.6824

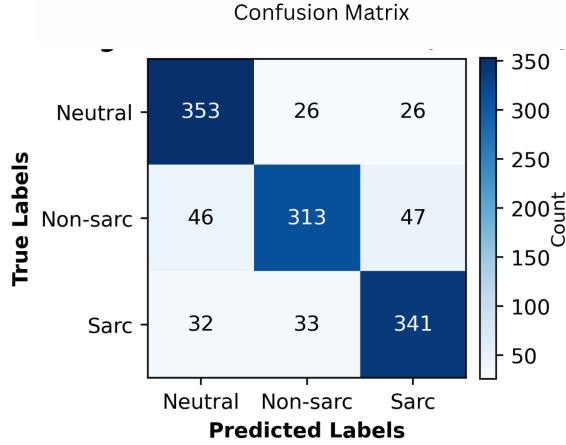


Fig. 6: Confusion matrix for BanglaSarc3 [3].

The ternary confusion matrix in Fig. 6 shows clear separability among sarcastic, non-sarcastic, and neutral instances, with slight overlap between sarcastic and non-sarcastic classes. Adapting bilingual Bangla–English inputs with R-Drop regularization led to a 3.1% macro F1-score improvement over the monolingual baseline.

C. Combined Comparison with Prior Work

To evaluate performance significantly against existing methods, we compared our results with GAN-BERT and several transformer baselines shown in the TABLE IV. The proposed supervised model achieved consistent improvements across all labeled data ratios.

TABLE IV: Comparison of BanglaBERT (FGM Fine Tuned) with Variable transformers and GAN-BERT models on BenSarc [2]

Model	3%	30%	40%	50%
Epochs	8	8	4	4
BanglaBERT (+FGM)	68.08	76.44	77.11	77.71
Bangla-BERT (GAN) [10]	72.9	76.4	77.0	77.1
Bangla-BERT (base) [10]	62.7	67.3	71.4	71.6
Indic-Transformer (Bengali-BERT) [10]	70.4	75.6	75.1	76.1
mBERT (cased) [10]	50.2	62.3	70.1	70.5
XLM-RoBERTa [10]	50.1	51.2	52.2	68.9

The supervised BanglaBERT (with FGM) achieved 77.71% accuracy at 50% labels, surpassing GAN-BERT by 0.61%, BanglaBERT-base by 6.11%, and XLM-RoBERTa by 8.81%. Even with only 3% supervision, it closely matched GAN-BERT’s 72.9%, demonstrating strong robustness without

semi-supervised learning. The proposed framework consistently produced 2–3% higher F1-score across both datasets, while integrating R-Drop and adversarial regularization improved ternary performance by nearly 2%. Overall, supervised BanglaBERT fine-tuned with FGM exhibited strong generalization, outperforming prior transformer and GAN-based models for Bengali sarcasm detection.

VI. CONCLUSION

Sarcasm detection in the Bengali language has gained significant importance with the rise of social media content. This study introduced a fully supervised BanglaBERT framework enhanced with Fast Gradient Method (FGM) adversarial learning, selective layer freezing, and R-Drop regularization to improve robustness and generalization. Experiments on BenSarc [2] and BanglaSarc3 [3] showed that the proposed model outperformed existing transformer and semi-supervised architectures. On BenSarc, the model achieved 89.00% accuracy and 0.890 F1-score, surpassing GAN-BERT by 0.6% in accuracy and 2–3% in F1-score. For the ternary BanglaSarc3 dataset, it attained 82.74% accuracy and 0.8275 macro F1-score, setting a new benchmark for Bengali sarcasm detection. The six-layer freezing strategy provided 2–7% performance gains by balancing adaptation and generalization. Future research can extend this framework to multi-domain and multimodal sarcasm detection, incorporating hybrid or semi-supervised approaches for greater robustness and scalability in Bengali NLP.

REFERENCES

- [1] T. S. Apon, R. Anan, E. A. Modhu, A. Suter, I. J. Sneha, and M. G. R. Alam, “BanglaSarc: A dataset for sarcasm detection,” *arXiv preprint*, arXiv:2209.13461, 2022.
- [2] S. K. Lora, G. M. Shahariar, T. Nazmin, N. N. Rahman, R. Rahman, M. Bhuiyan, and F. M. Shah, “Ben-Sarc: A self-annotated corpus for sarcasm detection from Bengali social media comments and its baseline evaluation,” *Natural Language Processing*, Cambridge University Press, 2024.
- [3] S. Biswas, M. M. R. Zahid, M. T. Rabeya, M. Abedin, and M. H. I. Bijoy, “BanglaSarc3: A benchmark dataset for Bangla sarcasm detection from social media to advance Bangla NLP,” *Data in Brief*, 2025.
- [4] R. Anan, T. S. Apon, Z. T. Hossain, E. A. Modhu, S. Mondal, and M. G. R. Alam, “Interpretable Bangla sarcasm detection using BERT and explainable AI,” in *Proc. IEEE 13th Annual CCWC*, 2023, pp. 1272–1278. doi: 10.1109/CCWC57344.2023.10094051
- [5] S. Khanom, F. Arefin, M. M. Hassan, and T. Sultana, “A comparative analysis of sarcasm detection in Bangla texts using several AI approaches,” in *Proceedings of Springer Book Chapter*, 2025.
- [6] A. Vaswani *et al.*, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019.
- [8] A. Bhattacharjee *et al.*, “BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding,” in *Findings of NAACL*, 2022, pp. 1318–1327.
- [9] T. Miyato, A. M. Dai, and I. Goodfellow, “Adversarial training methods for semi-supervised text classification,” *arXiv preprint*, arXiv:1605.07725, 2017.
- [10] S. K. Lora, I. Jahan, R. Hussain, R. Shahriyar, and A. B. M. A. A. Islam, “A transformer-based generative adversarial learning to detect sarcasm from Bengali text with correct classification of confusing text,” *Heliyon*, vol. 9, no. 11, p. e22531, 2023. doi: 10.1016/j.heliyon.2023.e22531